Gabby Martinez

Professor Duggan

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A Disconnected T: Who has access to Boston's public transportation?

Abstract

It is crucial to a well functioning and equitable society to allow all people equal access to everyday places like work, schools, grocery stores, and more. With public transportation, which is predominately used in large cities in the United States, people are granted this access, especially if they cannot provide their own transportation. However, public transportation is known to have discrepancies between populations of different races and socioeconomic status. This study aims to analyze these issues as they exist in Boston, Massachusetts, and explore how Boston's public transportation serves its communities and how it provides access to different populations and neighborhoods. Using multiple MBTA datasets regarding ridership information and T quality, as well as Boston demographic data, this paper gets closer to uncovering the situation in Boston. It finds that the racial and economic gaps do persist in Boston as well, and that the majority of riders are unhappy with their experiences on public transportation. Finally, these findings highlight the need for a reorganization of Boston's public transportation as well as possible new additions in modes of transportation, free access to the T, and rethinking how to serve the communities who need it the most.

Introduction

If you are a resident of a city in the United States, you probably have a few words to say about your local public transportation system. Common complaints include it's physically

unpleasant to be in (smells bad, loud, hot and no air conditioning, etc.), it takes too long to get to your desired destination, it's always undergoing construction or has lines closed, and it's too expensive. Maybe you don't even take your local public transportation because these issues are so bad or it serves you so poorly that it's not worth the effort. While frustrating enough that one would resort to other methods of transportation than subways and buses, like personal vehicles or rideshares, it is a privilege to be able to access these other modes of transportation. For many Americans, their reality is limited to only what they can access by way of public transport. Their only option is to take the bus that is 40 minutes late or the train that can at best get them a 20 minute walk from their destination. They have no choice but to bear the hardships of these public transportation nightmares.

In the U.S. 45% of people do not have access to public transportation, even though everyone from individuals to families to businesses could benefit from it (American Public Transportation Association). The term public transportation refers to many kinds of transportation modes and some that might not always come to mind, like ferries and water taxis, cable and street cars, commuter rails, and specific services for the elderly or people with disabilities, in addition to the stereotypical subway and bus systems. According to the American Public Transportation Association (APTA), "public transportation is a $79 billion dollar industry that employs more than 430,000 people" and $39 billion of that ends up going into the private sector. Upon discovering this I found myself feeling frustrated that a public good that is as crucial as this one is being turned into personal profit and at least partially controlled by private entities that will then dictate aspects of the public good. Additionally, the APTA states that home values that are near public transportation are up to 24% more expensive than those that are not close to public transportation. It is quite conflicting that public transportation is meant to be a

cheaper and more sustainable mode of transportation, but the fact that the cost of living near a public transit system is higher would offset the benefit of this cheap transportation.

Many studies have shown that there is a significant wealth gap when it comes to access to public transportation. The Institute for Transportation and Development Policy (ITDP) has shed light on the fact that the United States is a very car dependent country, more so than other developed nations. This has contributed to the lack of reliable public transportation in the country, and forced low income households to buy personal vehicles which understandably take up a larger portion of their income. The ITDP also states that "low-income communities of color are, as a result, disproportionately affected by a lack of access to reliable and accessible public transit services." Black and Hispanic workers are more likely to use public transportation when compared to White workers, but because of the previously mentioned high cost of living near public transportation, access to it is complicated and not easy.

The issue of transport inequality by factors such as race and income are explained more by Basav Sen from Inequality.org in "How the U.S. Transportation System Fuels Inequality." This article demonstrates how personal vehicle ownership is the main way of transportation in America. Additionally, just two-thirds of Black people in America have cars, compared to 86% of White people. Low income individuals are also less likely to own a vehicle. Again, why are the people who need access to public transportation also the highest disregarded population? Sen suggests it's a greed problem. The gas and oil industries have participated in heavy lobbying for the expansion and increase of roadways in the country which has been successful. Therefore, the country is focusing on economic gain for a handful of people rather than the collective good of all people because they are prioritizing roadways over public transportation.

The Reality in Boston

For riders in Boston, MA, the situation is no different. The public transportation system in Boston is known as the T and is operated by the Massachusetts Bay Transit Authority (MBTA). It includes a system of buses, four different subway train lines named by colors, a commuter rail line that shuffles people in and out of the city on an hourly schedule, and a poorly known water transport system. According to an analysis done by Transit Center, Black and Latino communities in Boston have the worst access to the T, measured by the ability to get to jobs, healthcare, and grocery stores, even though they have the highest ridership numbers. They also found that the income inequality issue when it comes to the T is quite simple: those with more money can reach more places. Lines like the commuter rail increase in price by the destination, and bus and subway tickets cost around $2.50 per ride. With a budget like $5 on transportation for a day, one can maybe just get to their destination and back, no switches or reroutes, which is difficult when it requires multiple trains or a bus and train to reach one's destination.

Studies have shown that Boston follows about the same trends in public transportation inequality that we see on the national level, that income inequality and race are big indicators of who has access and who doesn't. "An Exploration of Destination Access and Transportation Cost Analyses" by Betsy Harvey et al concludes that income inequality is the largest contributor to disparities in transportation and destination access. It also further breaks down destinations to access like school, work, healthcare, and parks, and discusses who can best get to each of these locations. Furthermore, research done by scholar Avery Heinz finds the similar story that people of color are most negatively affected by public transportation in Boston and again, emphasizes the prioritization of highways and gas and oil industries which so negatively affected the public transportation industry as it was left in the dust (Mastantuono).

While we know there are differences in how people can experience the T, there is not much light shed on where exactly these people are, the specific transportation they do and do not have access to, and an in depth look of why this might be the case for Boston specifically. Also, who has the best access to the T in regards to easy ability to reach desired destination, as well as the newest T cars, most frequently upkept lines, and general comfortable riding experience. My research aims to bring this information forward and provide the necessary insight on the MBTA that will assist the T in becoming more equitable and accessible for all.

Methods

In order to answer these questions I have analyzed publicly available MBTA data as well as Boston demographic data. When these datasets overlap, it should be able to tell a story about who has access to what and who is most consistently and effectively served by Boston's public transportation. There are many things that could be studied with the available MBTA datasets. One dataset I came across is a rider satisfaction survey. The survey asks questions along the lines of how one rates the MBTA overall and satisfaction with communication from the MBTA. This data can be used to determine accessibility to the T based on riders experiences and get a direct idea of how the T serves people from the people themselves. Are there people of certain demographics who rate the T higher than others? Is there a correlation with high ratings and specific T lines or areas? Another dataset for my research could be transit travel times. These show how much time it takes different modes of the T to reach its destination, testing the efficiency of it. Additionally, this information has been collected over multiple years, meaning one would be able to tell how the efficiency in the T has changed over time and exactly what parts of it have changed. Unfortunately, this dataset was not able to be used in this current research, but I hope to explore it in the future. Service alert data would also be helpful in

determining efficiency because it informs riders of delays and track shutdowns. It would also show what areas are affected the most by nonfunctional transportation, and comparing this to demographic data might reveal some answers. Lastly, the MBTA provides data on mode of transportation which shows how many people took a specific mode of transportation each month from 2016 to 2024. This would inform my research of what lines and routes are most used, and again, coupled with demographic data, one can infer who is using these lines the most. The research will be most effective if these datasets are analyzed together, because they can all provide insight on each other.

In addition to the MBTA data itself, I compiled resources of demographic data, including race and income specifically. The city of Boston publishes data regarding neighborhoods and people, including Boston Neighborhood Data on Analyze Boston and 2020 Census Data. The source is extremely reliable and breaks down the population of each neighborhood by race and ethnicity. It also includes an understanding of age in each neighborhood as well, sorting the data into buckets of 0-17 and 18+. Finally, the data has information on healthcare facilities, university housing numbers, and correctional facilities which will provide insight on the kinds of people who live in these neighborhoods and if the city has structured it, and their public transportation, this way for a reason. By overlapping these datasets correlations may be found for the questions I want to answer regarding who has appropriate access to the T. Finally, for the analysis, the data has been explored using Python.

Data Analysis Findings

Boston MBTA Data

The first thing the data was used to explore was what mode of transportation of the T is used the most. Using the MBTA Ridership dataset, it was found that from 2016 to 2024 the Bus

has had the highest number of riders every year. Below is the dictionary that represents the number of riders for the Bus for each year:

> Highest ridership for each year: {'2016': ['Bus', 102290543.0], '2017': ['Bus', 102307370.0], '2018': ['Bus', 103520208.0], '2019': ['Bus', 103654773.0], '2020': ['Bus', 54064668.0], '2021': ['Bus', 58896756.0], '2022': ['Bus', 72109180.0], '2023': ['Bus', 78461729.0], '2024': ['Bus', 55203564.0]}

In contrast to that, the Silver Line had the least amount of riders per year. However, this excluded the Boats, Ferry, and The RIDE, as those all had lower ridership numbers. I decided to exclude those because they are already known to be the least used methods within the Boston community and I wanted to see between different modes of buses and trains which was the least used. Figure 1 will also demonstrate the stark difference between ridership for these modes. Below are the statistics for the Silver Line ridership for each year:

> Lowest ridership for each year: {'2016': ['Silver Line', 9298915.0], '2017': ['Silver Line', 9339789.0], '2018': ['Silver Line', 11051561.0], '2019': ['Silver Line', 11752348.0], '2020': ['Silver Line', 5587981.0], '2021': ['Silver Line', 6161570.0], '2022': ['Silver Line', 8205614.0], '2023': ['Silver Line', 9361781.0], '2024': ['Silver Line', 6610584.0]}

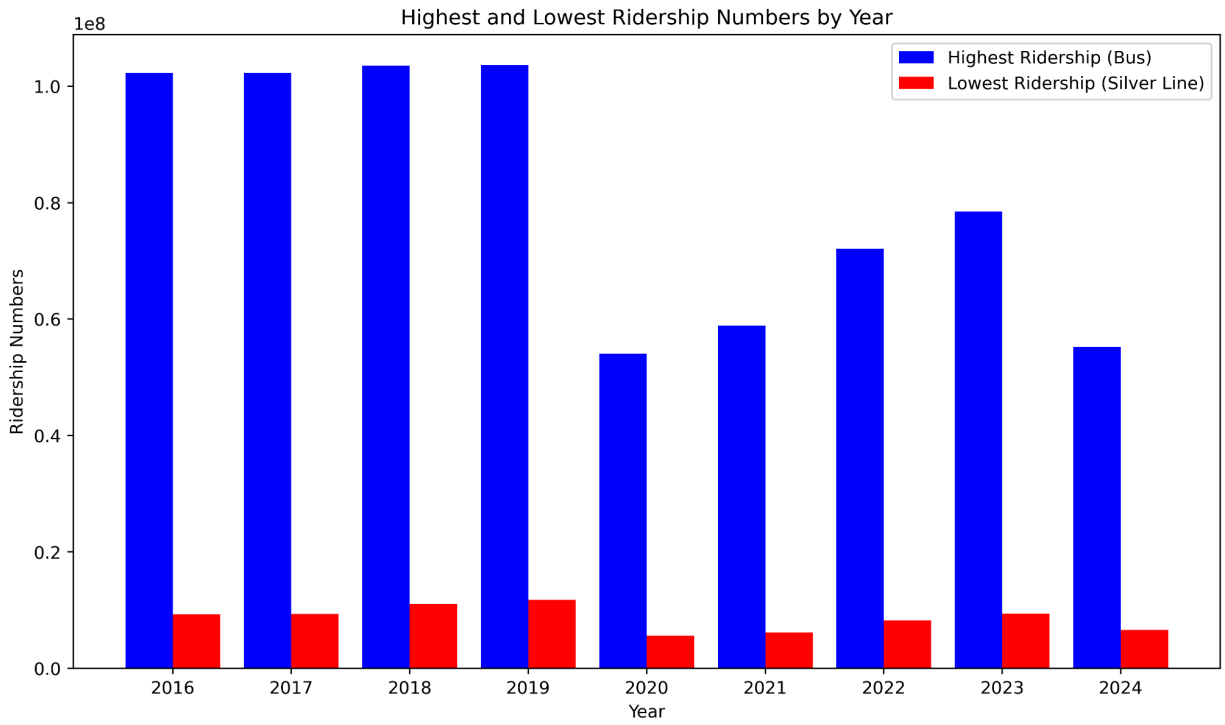


Figure 1: Shows the highest and lowest ridership numbers from 2016-2024. The blue bar represents the Bus, with the highest ridership, and the red bar represents the Silver Line, which has the lowest ridership. The ridership numbers are log scaled for better visualization purposes, as they differ drastically. Source: MBTA Monthly Ridership by Mode ("MBTA Monthly Ridership by Mode")

When analyzing this information with background knowledge of the T and where it serves, one can conclude that people who live in low income and racially diverse areas, where the predominant mode of transportation is the Bus, use public transportation in Boston the most. On the other hand, the Silver Line serves wealthier communities like Seaport, as well as the airport, and is used the least frequently because it is not needed as much there.

Using the publicly available MBTA data on service alert information, I was able to find that the Bus is also the mode that experiences the highest number of service alerts each year, from 2016 to 2023.

Mode with most service alerts: {'2000': ['Bus', 2, 'snow'], '2019': ['Bus', 30803, 'traffic'], '2020': ['Bus', 6325, 'traffic'], '2021': ['Bus', 10376, 'traffic'], '2022': ['Bus', 15682, 'traffic'], '2023': ['Bus', 8823, 'traffic']}

As you can see here, the most frequent cause for the service alerts each year is traffic. While traffic is always going to be an issue with ground transportation such as a bus, it does not distract from the fact that it still heavily affects these communities whose only option is to take the Bus.

Additionally, throughout the entire dataset there are 157,786 delays on record that took up to 20 minutes or more. That is a total of at least 6 years worth of waiting due to delays, within the timeframe of the data. That is a substantial amount of time to wait for public transportation. For each delay, the MBTA provides a scale, from 1 to 10, that determines how bad the delay is. A code of 3 indicates the delay was up to 10 minutes, a 5 is up to 20 minutes, and a 9 indicates about an hour. The Bus has an average severity code of 5.37, which is the highest of all the modes of transportation. The full dictionary can be found here:

Average severity code per mode: {'Access': 2.98, 'Bus': 5.37, 'Commuter Rail': 3.94, 'Subway': 4.02, 'Boat': 3.88, 'Systemwide': 3.65}

Most common delay reasons per mode: {'Access': 'maintenance', 'Bus': 'traffic', 'Commuter Rail': 'track work', 'Subway': 'disabled train', 'Boat': 'mechanical problem', 'Systemwide': 'holiday'}

I also included above an analysis of the most common delay reasons per mode, where again the Bus is most affected by traffic. In addition to that, Figure 2 is a graph of the average severity code for each mode, to better understand the data.

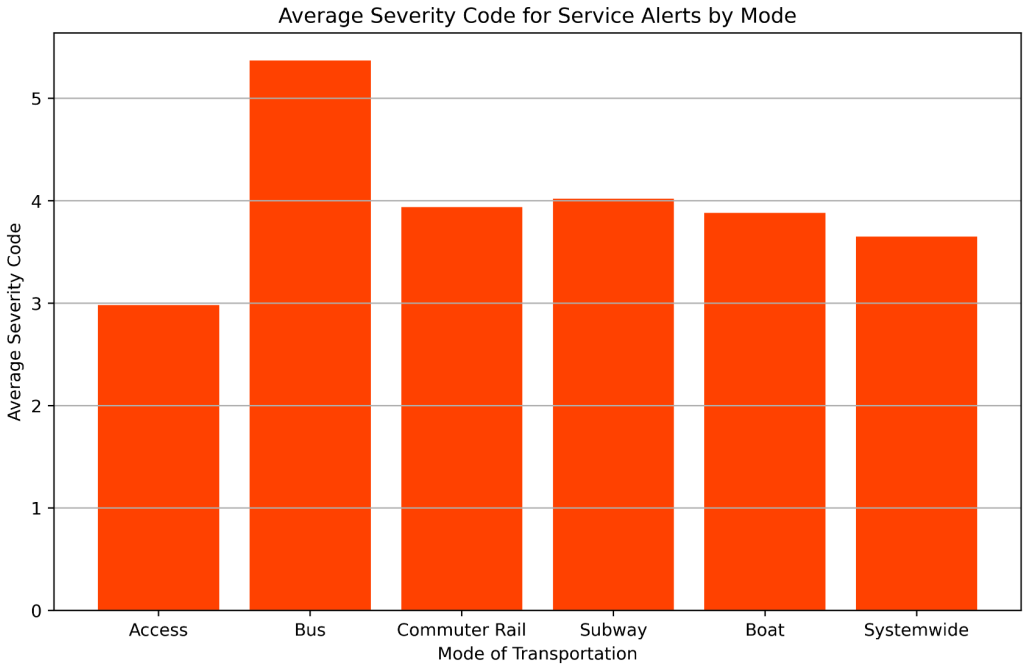


Figure 2: Shows the average severity code for each mode of transportation. Source: MBTA Service Alerts ("MBTA Service Alerts")

The final thing I wanted to examine with the available MBTA data was rider's satisfaction levels. While this could not tell me what riders were more or less satisfied, I was able to explore general trends with the data. Figure 3 shows for each year of collected data the average percent of respondents that were satisfied with the T, dissatisfied with the T, or neutral. The original survey used a 7-point scale, where 3 answers demonstrated being unsatisfied and 3 demonstrated satisfaction. I added the percentages from the 3 dissatisfied and 3 satisfied for each month, and then averaged the months together for every year.

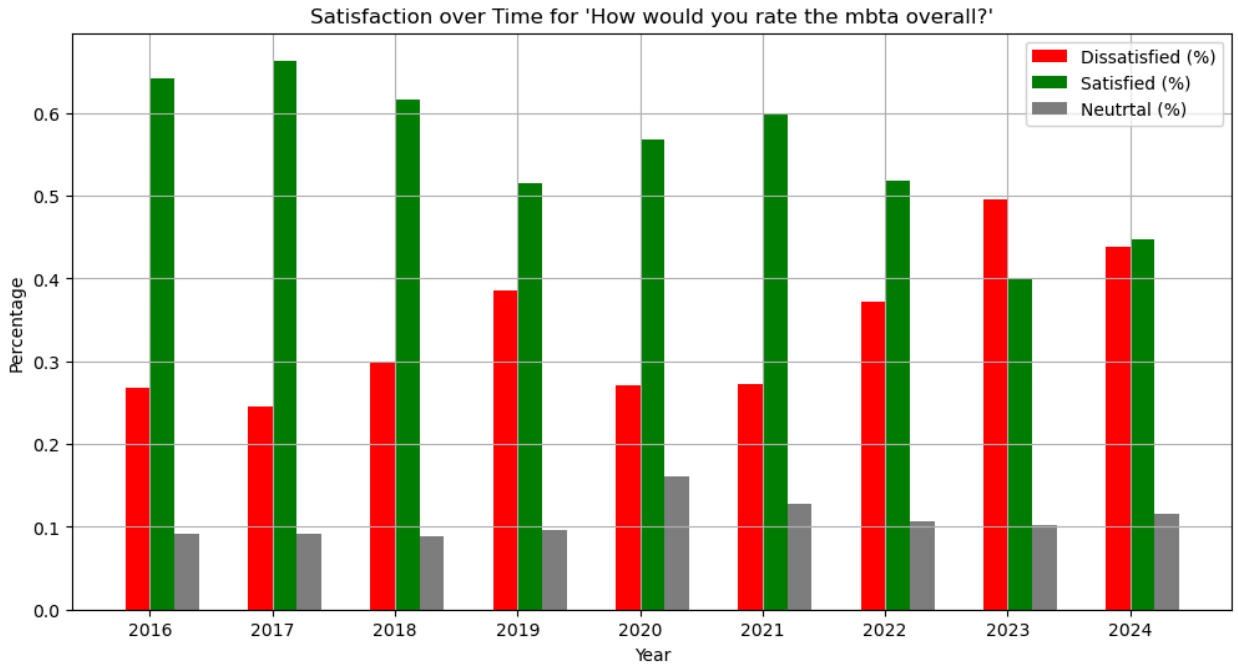


Figure 3: Percentage of T riders who expressed satisfaction and dissatisfaction with the T over time. Source: MBTA Service Customer Satisfaction Survey ("MBTA Service Customer Satisfaction")

The graph shows that overall satisfaction with the T has been decreasing, and in 2023 dissatisfactory responses surpassed satisfactory ones since the data became available. It is possible that this will be the case again for 2024 as the year comes to an end. Exploring this data further enforces the fact that riders and users of Boston's public transportation systems are unhappy with their experiences and suggests change must be made somewhere.

The main takeaway from these analyses of various Boston MBTA datasets is that the Bus is the worst mode of transportation and Bostonians are unhappy with their T experiences. The Bus has consistently had the most issues and the highest ridership. Is it a coincidence, or is it speaking to a larger economic and racial divide in Boston? Next I turn to analyzing Boston neighborhood data to get a better idea of this question.

Boston Demographics Data

First and foremost, it is crucial to understand the racial makeup of Boston. For that purpose, Figure 4 demonstrates the percentage of the population that is White, Black, Asian, and Hispanic for each neighborhood. The graph shows that fifteen of the demonstrated neighborhoods are at least 50% White. This also seems to correlate with the neighborhoods that have more transit options, like Back Bay and the North End. In contrast, Hyde Park, East Boston, and Roxbury have larger populations of Hispanic and Black people and simultaneously one, or no, subway lines to use.

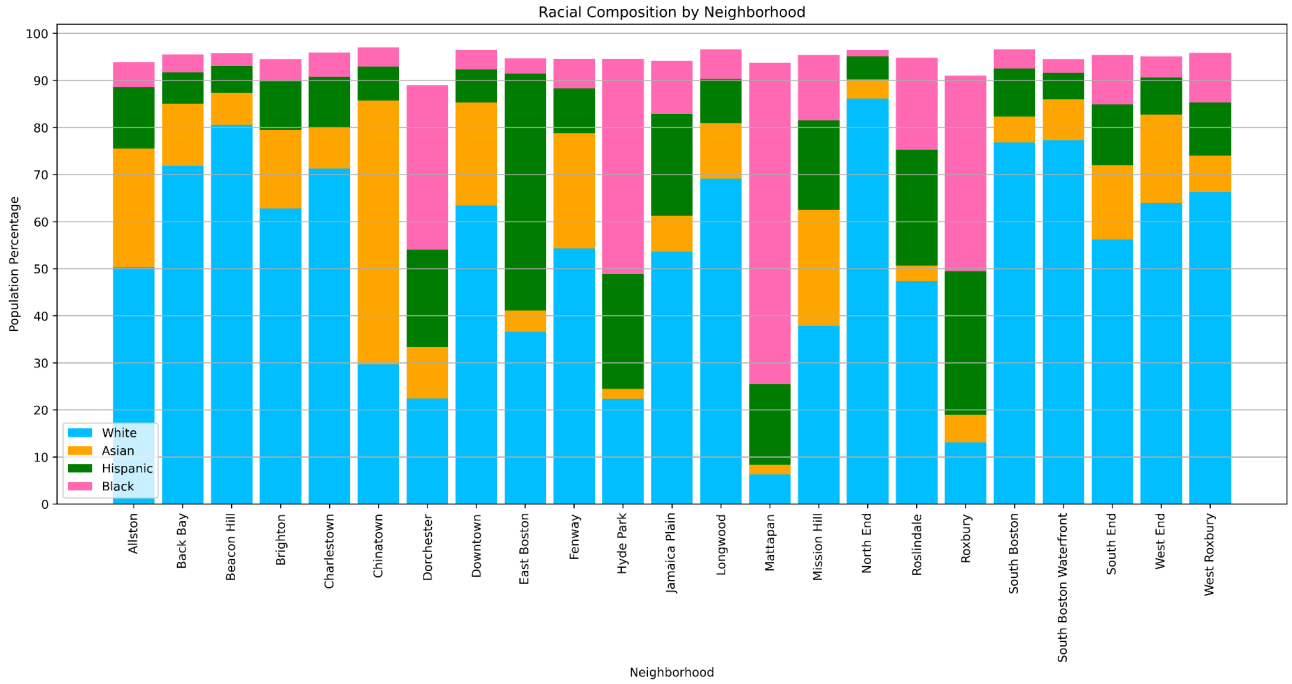


Figure 4: Demonstrates the racial composition of each Boston neighborhood. Includes percentages of White, Asian, Hispanic, and Black populations. Does not include individuals that indicated mixed race. Source: Boston Census Data ("2020 Census for Boston - Boston Neighborhood Data - Analyze Boston")

Additionally, data regarding median household income in Boston shows that White neighborhoods seem to have higher incomes. For example, Beacon Hill has a median income of $118,125, while Roxbury is just under $35,000 ("Income Household Median in Boston, MA by Zip Code. Map and Data. Printable Map. | Simplemaps.com"). The information presented here is

in line with that of scholars and writers previously mentioned who shed light on the wealthier and whiter populations having better access to public transportation.

Finally, I analyzed the data regarding prison population and student population for each neighborhood. This data comes from 2020 census data on Boston neighborhoods. Additionally, it is unclear what exactly a correctional facility is, but can be useful in understanding the population nonetheless. The neighborhoods with adult and juvenile correctional facilities, and how many, can be found below.

> Neighborhoods with adult correctional facilities: {'Fenway': 53, 'Mattapan': 86, 'Roxbury': 726, 'South End': 16, 'West End': 34}
>
> Neighborhoods with juvenile correctional facilities: {'Brighton': 13, 'Dorchester': 286, 'East Boston': 12, 'Jamaica Plain': 90, 'Mattapan': 75, 'Mission Hill': 4, 'Roslindale': 9, 'Roxbury': 160, 'South Boston': 56}

Roxbury has an outstanding 886 facilities between both categories, more than any other Boston neighborhood. Some of these neighborhoods have decent access to the T, like Fenway, but most of them do not appear to be in the City of Boston's most important areas for public transportation infrastructure. Upon further inspection, one could determine if there are less/worse public transportation options in these areas with higher prison populations because of societal biases against these communities.

For the areas with higher student populations, I hypothesized that they would have better access to the T because of the power of wealthy institutions in Boston. I am using the number of student housing units as a proxy for the number of students in a given neighborhood, and therefore also an indicator of the education level of that population. The neighborhoods with student accommodations are as follows:

Neighborhoods with student housing: {'Allston': 3214, 'Back Bay': 1487, 'Brighton': 3493, 'Chinatown': 443, 'Downtown': 3283, 'Fenway': 14960, 'Hyde Park': 64, 'Jamaica Plain': 160, 'Longwood': 4624, 'Mission Hill': 1502, 'Roxbury': 2520}

A few of the neighborhoods here overlap with the neighborhoods with correctional facilities, but the number of student housing units far outweighs the number of correctional facilities. Areas with a lot of student housing, like Back Bay, Downtown, and Fenway, have decent access to the T as they can be reached by all or most lines of the Green Line, the Orange Line, and easy access to the Red Line. Areas with less student housing, such as Jamaica Plain and Hyde Park, do not have the same easy access.

Limitations

As this research developed, there were some limitations that arose. First and foremost, I was not able to filter all the datasets I used by demographic information, which means I was not able to find a direct correlation between factors. For example, the MBTA Rider Satisfaction Survey data does not include individual rider information, which is most likely for ethical reasons, and also doesn't include any questions about what line(s) the rider takes most frequently, so I cannot make any specific claims or conclusions with that dataset. Generalizations can be made with it, and were, but by itself is not very telling of specific neighborhood rider information. Additionally, the datasets that were analyzed did not always match up with the same time, meaning one dataset might have information for some years and other datasets for other years. While they can still be analyzed and discussed together, they can't be directly connected to each other because the information from them was not gathered at the same time. Additionally, demographic data is sometimes hard to come by for ethical and privacy reasons of the individuals. All of the data is also subject to data collection biases that should be taken into account and considered when analyzing the results. These may include respondent biases, data

collection errors, or misinterpretations due to lacking information. Finally, I would've liked to work with a dataset that included direct neighborhood to line/mode of transportation information, as that would make it better to understand direct correlations and run regression analyses.

One final major limitation I'd like to note as well is regarding the way public transportation in Boston is used. As a current resident in Boston, I am highly aware of the ways one can access some modes of transportation here for free by fare evading. The Bus is the hardest mode to fare evade on. So, if ridership numbers were counted using how many times a fare was paid, the data could be skewed because it is not counting the ridership of fare evaders. Fare evading is very common on Green Line street level stops, and also common on the subway by sneaking behind another rider or opening the gates from the opposite side for another to go through. While the data coming directly from the MBTA is reliable, it may be excluding information and definitely is not as clear as it could be to be in terms of how the data was collected and put together.

Conclusion

I had hypothesized that the data will show, with my operationalization of accessibility to the T, that those in areas of lower income and more racial diversity will experience a worse T and less accessibility to other areas of Boston. This means they will experience things such as slower transit times, more track work and delays, overall slower transportation modes (like the Bus), and limited availability of public transportation options. With the data I was actually able to analyze, it was found that the Bus has the highest ridership numbers, serves the minority population the most, and has the highest number of delays as well as severity of said delays. It was also found that overall satisfaction with the MBTA is decreasing, suggesting that Boston's

public transportation system is not keeping up with the needs of Bostonians in everyday life today.

One solution to improve access to the T and to other areas of Boston is to provide fare-free access to public transportation for all individuals. The MBTA has been testing this out with some bus routes already, the 28, 23 and 29, which are routes that specifically go through Roxbury and Dorchester ("Fare-Free Program for Routes 23, 28, and 29 | Projects | MBTA"). The project is lasting 4 years, and we are currently in the middle of it. The main goal of this is to allow low income individuals better opportunities to travel. If all public transportation in Boston became free, everyone would be able to access every part of the city with more ease, however, it does not fix the issue that there are fewer direct routes between certain neighborhoods which still makes commuting difficult for some and divides the city.

To combat the physical lack of infrastructure and public transportation in certain areas of Boston, the city needs to take a good look at restructuring and adding or changing lines and routes. The City of Boston recently announced a new Green Line train car that will be hitting the tracks in the next few years and upon attending an event regarding this, I learned that the new cars will be placed on the D line. When asked why the E Line has consistently had the oldest cars, and also happens to serve one of the lowest income communities of the entire Green Line, I was told it was due to safety issues with some of the stops on the route. These are stops that let the riders off right in the middle of the road and do not have any measures to keep passengers safe. It is crucial for Boston to fix issues like this as a start to improving access to public transportation. Next, the city should update all the train cars and buses to ensure a good quality standard and minimize the amount of delays and issues caused by old vehicles. This would greatly improve efficiency and travel time. Also to improve travel time, a restructuring of some

existing tracks would be beneficial. Some of the stops have sharp bends and turns that force the trains to move very slowly around them, taking longer and creating a higher risk for accidents.

Once the city is able to update all existing transportation forms, I suggest they open new lines that can connect previously unconnected neighborhoods, specifically ones of lower income or higher minority percentage to those of higher income and lower minority percentage. As of right now, there are Bus routes such as the 66 and the 1 that connect areas like Mission Hill and Roxbury to areas of Cambridge, but as we've seen the buses can be unreliable and difficult to manage. Instead, the city should adopt new train lines or efficient bus options by enforcing bus only lanes. There are many more ways the City of Boston and the MBTA can look to connect communities and create a more efficient and comfortable experience on public transportation. For now, this paper serves as a stepping stone into those ideas and an exploratory analysis of the problems facing Boston's public transportation today and in its recent history.

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